

The origin of contraversive pushing

Evidence for a second graviceptive system in humans

H.-O. Karnath, MD, PhD; S. Ferber, PhD; and J. Dichgans, MD

Article abstract—*Background:* Stroke patients may exhibit the peculiar behavior of actively pushing away from the nonhemiparetic side, leading to lateral postural imbalance and a tendency to fall toward the paralyzed side. This phenomenon has been called the “pusher syndrome.” *Objective:* The current study analyzes the mechanism leading to contraversive pushing. *Methods:* The subjective postural vertical (SPV) and subjective visual vertical (SVV) were determined in five consecutively admitted patients with severe contraversive pushing and in controls. Whereas adjustment of the SPV reflects the perceived upright orientation of the body, the SVV provides a sensitive and direction-specific measurement of peripheral and central vestibular dysfunction. *Results:* The deficit leading to contraversive pushing is an altered perception of the body's orientation in relation to gravity. Pusher patients experience their body as oriented “upright” when it is tilted 18° to the nonhemiparetic, ipsilesional side. In contrast, perception of the SVV was undisturbed. *Conclusions:* A separate pathway seems to be present in humans for sensing the orientation of gravity apart from the one for orientation perception of the visual world. This second graviceptive system decisively contributes to humans' control of upright body posture. Contraversive pushing occurring after stroke lesions may represent the behavioral correlate of a disturbed neural representation of this system.

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Several studies have investigated the ability of humans to determine upright body position. Even in the absence of visual cues, healthy subjects show no apparent difficulties when asked to orient their body earth-vertical.^{1–3} Still, it is not well understood how the brain perceives verticality of the body and controls upright body posture.⁴ We know that a lesion of the human so-called vestibular cortex⁵ (i.e., the posterior insula) may lead to vestibular dysfunction in terms of a perceptual tilt of the visual vertical, but such lesions do *not* induce dyscontrol of body posture (for example, with falling to one side). Recent studies have investigated the perception of postural upright in patients with acute peripheral and various central vestibular disorders, as well as in normal subjects under vestibular or optokinetic stimulation.^{6,7} Surprisingly, they found that despite concurrent perception of vertigo, vection, and tilt of the subjective visual vertical (SVV), patients with, for example, vestibular neuritis or patients who underwent vestibular neurectomy, as well as healthy subjects under stimulation, show a stable, *undisturbed* perception and performance of body verticality.

Nevertheless, we know that an acute loss of postural balance may occur after brain damage. This clinical phenomenon has been termed the “pusher syndrome.”⁸ Whereas most patients with hemiparesis have good trunk balance soon after the stroke,

these latter patients, even when sitting, exhibit the peculiar behavior of *actively* pushing away from the nonparetic side, leading to postural imbalance and a tendency to fall toward the paralyzed side. When sitting or standing, they lean toward the hemiparetic side and resist any attempt to passively correct their tilted body posture. They use the nonparetic arm to *actively resist* attempts of correction toward the earth-vertical upright orientation and report the impression of lateral instability and the fear of falling toward the nonparalyzed side. In contrast, these patients show no fear when their active pushing leads to an unstable, tilted body position toward the paretic, contralesional side.

Although it has long been known that hemiparetic patients may show some postural imbalance^{9–14} or (due to paresis) may lose lateral balance, falling toward the paralyzed side,^{15–18} Davies⁸ first discovered that certain patients exhibit an *active pushing away* from the nonparalyzed side. This active pushing with the nonparetic extremities also distinguishes these patients from those with lateropulsion in Wallenberg syndrome^{19,20} and from those with astasia after thalamic lesions.²¹ When asked to sit up, patients in the latter group use the nonparetic arm to grasp, for example, the side rail of the bed “to *pull* themselves up.”²¹ Pusher patients do exactly the opposite. They extend this arm in the same situation and use it to *actively push away*, toward the paretic side.

From the Departments of Cognitive and General Neurology, University of Tübingen, Germany.

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Address correspondence and reprint requests to Dr. H.-O. Karnath, Department of Cognitive Neurology, University of Tübingen, Hoppe-Seyler-Str. 3, D-72076 Tübingen, Germany; e-mail: Karnath@uni-tuebingen.de

Table 1 Clinical assessment scale for contraversive pushing*

	Sitting	Standing
(A) Posture (symmetry of spontaneous posture)		
Score 1 = severe contraversive tilt with falling to the contralesional side	☐	☐
Score 0.75 = severe contraversive tilt without falling	☐	☐
Score 0.25 = mild contraversive tilt without falling	☐	☐
Score 0 = no tilt / upright body orientation	☐	☐
Total (max = 2):		
(B) Extension (use of the arm/leg to extend the area of physical contact to the ground)		
Score 1 = performed already in rest	☐	☐
Score 0.5 = performed not until position is changed	☐	☐
Score 0 = no extension	☐	☐
Total (max = 2):		
(C) Resistance (resistance to passive correction of posture to an upright position)		
Score 1 = resistance is shown	☐	☐
Score 0 = resistance is not shown	☐	☐
Total (max = 2):		

* Translated from Karnath H-O, Brötz D, Götz A. Clinical symptoms, origin, and therapy of the "pusher syndrome." *Nervenarzt*. In press.

Max = maximum.

Based on her clinical observations from physiotherapy, Davies⁸ assumed that the contraversive pushing is part of a syndrome, the "pusher syndrome," encompassing spatial neglect and anosognosia and that it is predominantly associated with lesions of the right hemisphere. A systematic investigation of her assumption in a large sample of acute stroke patients with hemiparesis²² confirmed the existence of pushing but revealed no evidence for a *syndrome*, that is, the regular combination of pushing with spatial neglect or anosognosia. No differences were observed in the incidence of spatial neglect, anosognosia, aphasia, or apraxia in patients with or without contraversive pushing.²² Moreover, the authors found that the disturbance occurred with equal frequency with either left or right brain damage. Thus, contraversive pushing does not appear to be caused by other neuropsychological symptoms. Instead, the patients' subjective impression of becoming "unstable," combined with their fear of "falling" when trying to passively correct the contraversively tilted body axis, led us to assume that these patients might experience a misperception of body orientation in relation to gravity. The current study supports this hypothesis and provides the first evidence for the mechanism leading to the contraversive pushing.

Methods. *Subjects.* Fifteen subjects were examined. In five consecutively admitted patients with hemiparesis and severe contraversive pushing but no expressive or receptive language disturbances or visual field defects, we de-

termined the subjective postural vertical (SPV) and the SVV. Whereas adjustment of the SPV reflects the perceived upright orientation of the body, the SVV provides a sensitive and direction-specific measurement of peripheral and central vestibular dysfunction. (Patients with left hemisphere lesions who had contraversive pushing and aphasia were excluded because our experimental procedure did not allow for any communication problems.)

Contraversive pushing was assessed using the standardized Scale for Contraversive Pushing (SCP).²³ A translation of this scale is given in table 1. Based on Davies' criteria,⁸ the SCP assesses the following: 1) symmetry of spontaneous posture while sitting and standing, 2) the use of the arm or the leg to extend the area of physical contact to the ground while sitting and standing, and 3) resistance to passive correction of posture while sitting and standing. Patients were scored as having contraversive pushing if all three criteria were present, reaching a total score of at least 1 (sitting plus standing) with respect to their spontaneous posture, a total score of at least 1 (sitting plus standing) concerning the use of the arm or leg to extend the area of physical contact, and a total score of at least 1 if they showed resistance to passive correction of posture while sitting and standing. The clinical and demographic details of all patients are given in table 2; their brain lesions are illustrated in figure 1.

The performance of the 5 patients with contraversive pushing was compared with 10 control subjects. Five of these subjects had unilateral right hemispheric stroke lesions and left-sided hemiparesis but did not show any evidence for contraversive pushing (see table 2, C1-5). Because all five of the pusher patients had spatial neglect

Table 2 Demographic and clinical data for the right brain-damaged patients with contraversive pushing (P1–5), for the control patients with right-sided brain lesions and neglect (C1–4), and for the control patient with hemianesthesia after a right thalamic lesion (C5)

Patient	Sex/age (y)	Etiology	Days after onset	Hemiparesis	Hemihypesthesia	SCP*			Spatial neglect	Letter cancellation†		Bells Test‡		Copying□	
						Post	Ext	Resis		Left	Right	Left	Right	Left	Right
P1	F/79	Hem	5	+	+	2	2	2	+	0	16	0	8	(–)	+
P2	F/69	Hem	24	Hemiplegia		2	1	2	+	0	11	0	6	–	+
P3	F/74	Hem	10	+	+	2	2	2	+	0	4	0	5	(–)	+
P4	M/77	Infarct	13	+	+	2	2	2	+	0	6	0	9	(–)	(–)
P5	M/69	Infarct	14	+	+	2	2	2	+	6	25	0	14	–	+
C1	F/31	Infarct	7	+	+	0	0	0	+	17	28	7	12	(–)	+
C2	M/58	Infarct	34	+	+	0	0	0	+	0	15	0	10	–	(–)
C3	M/65	Hem	500	+	+	0	0	0	+	0	6	0	4	(–)	+
C4	F/62	Hem	7	+	+	0	0	0	+	0	11	0	2	–	+
C5	M/51	Hem	5	+	Hemianesthesia	0	0	0	No	30	30	13	15	+	+

* Clinical Assessment Scale for Contraversive Pushing (SCP)²³ assesses 1) symmetry of spontaneous posture (Post) while sitting and standing (max, 2), 2) the use of the arm or the leg to extend (Ext) the area of physical contact to the ground while sitting and standing (max, 2), and 3) resistance (Resis) to passive correction of posture while sitting and standing (max, 2).

† Letter cancellation³³ presents 60 target letters “A” on a 21 × 29.7 cm large sheet of paper (30 on its right half, 30 on its left) interspersed among various distractor items in a random array of letters. The subjects are asked to mark all of the letters “A” on the whole page. The maximum score is 60.

‡ Bells Test³⁴ consists of seven columns each containing five targets (bells) and 40 distractors. Three of the seven columns are on the left side of a 21 × 29.7 cm large sheet of paper (=15 targets); one is in the middle and three are on the right side (=15 targets). The patient is asked to cross out all of the bells; the maximum score is 30.

□ Copying task is a test where patients are asked to copy a complex multiobject scene consisting of four elements (a fence, a car, a house, and a tree), two in each half of a horizontally oriented 21 × 29.7 cm large sheet of paper: –, omission of at least one whole object; (–), omission of at least one left-sided feature of a figure; +, copying without omissions.

Hem = hemorrhage.

in addition to contraversive pushing, four of these brain-damaged controls (C1–4) were selected for their spatial neglect. The fifth brain-damaged control patient (C5) had no neglect but was chosen for a complete loss of somatosensory input from the paralyzed body side resulting from left-sided *hemianesthesia* after a right thalamic lesion. Because all of the pusher patients in this study had mild to moderate hemihypesthesia, this patient served to control for the pure influence of somatosensory loss on the perception of body posture. Five non-brain-damaged neurologic patients, aged 45 to 72 (median 59) years without a history of visual, vestibular, or somatosensory dysfunction, served as additional controls (three men, two women). All participants gave informed consent.

Apparatus and procedure. Subjects were seated upright on a motor-driven padded chair, immobilized by lateral stabilization. The height of the chair was adjusted so that the subjects' legs were freely hanging. Subjects were rotated in the frontal (roll) plane by the experimenter. After a random offset in the frontal plane of at least 35° either to the left or the right, the subjects verbally indicated in which direction they had to be repositioned to be upright. Repositionings were carried out by the experimenter in a stop-and-go procedure according to the subjects' verbal instructions. Subjects were required to indicate when they entered and left the upright position, which constituted the sector of subjective verticality. If subjects signaled that they had left the upright orientation and felt tilted again, they were rotated in that direction for an additional random interval of up to 15° before the direc-

tion of the movement changed. Ten repositionings were conducted, 5 in each direction of movement. (Because of clinical limitations, only six repositionings could be done with patient P2 when the eyes were occluded.) Subjects were not informed about their starting position and did not get any feedback about their performance. Rotations were performed 1) viewing the structured surroundings of the laboratory under normal lighting conditions, and 2) without visual cues, when the subjects' eyes were occluded. To determine an axis of subjective verticality, we calculated the mean of entering and leaving the upright position. The angular deviation from earth-vertical was defined as negative when the left ear was down.

In addition, the SVV was determined using a 7.5-cm long luminous rod presented in front of the upright sitting subject at eye level in complete darkness. Ten SVV adjustments were conducted, 5 in each direction of movement. From a random angular offset of at least 35° either to the left or the right, the subject was asked to set the rod so that it appeared to be earth-vertical. In parallel with the procedure used to determine the SPV, adjustments were made by the experimenter in a stop-and-go procedure following the verbal instructions of the subject. The angular deviations from earth-vertical were defined as positive for clockwise deviations and negative for counterclockwise deviations.

Results. As illustrated in figure 2, the 10 control patients with and without brain lesions showed no tilt of their SPV, either when their eyes had been occluded (mean 0.4°, SD 0.9°) or while viewing the surroundings (mean

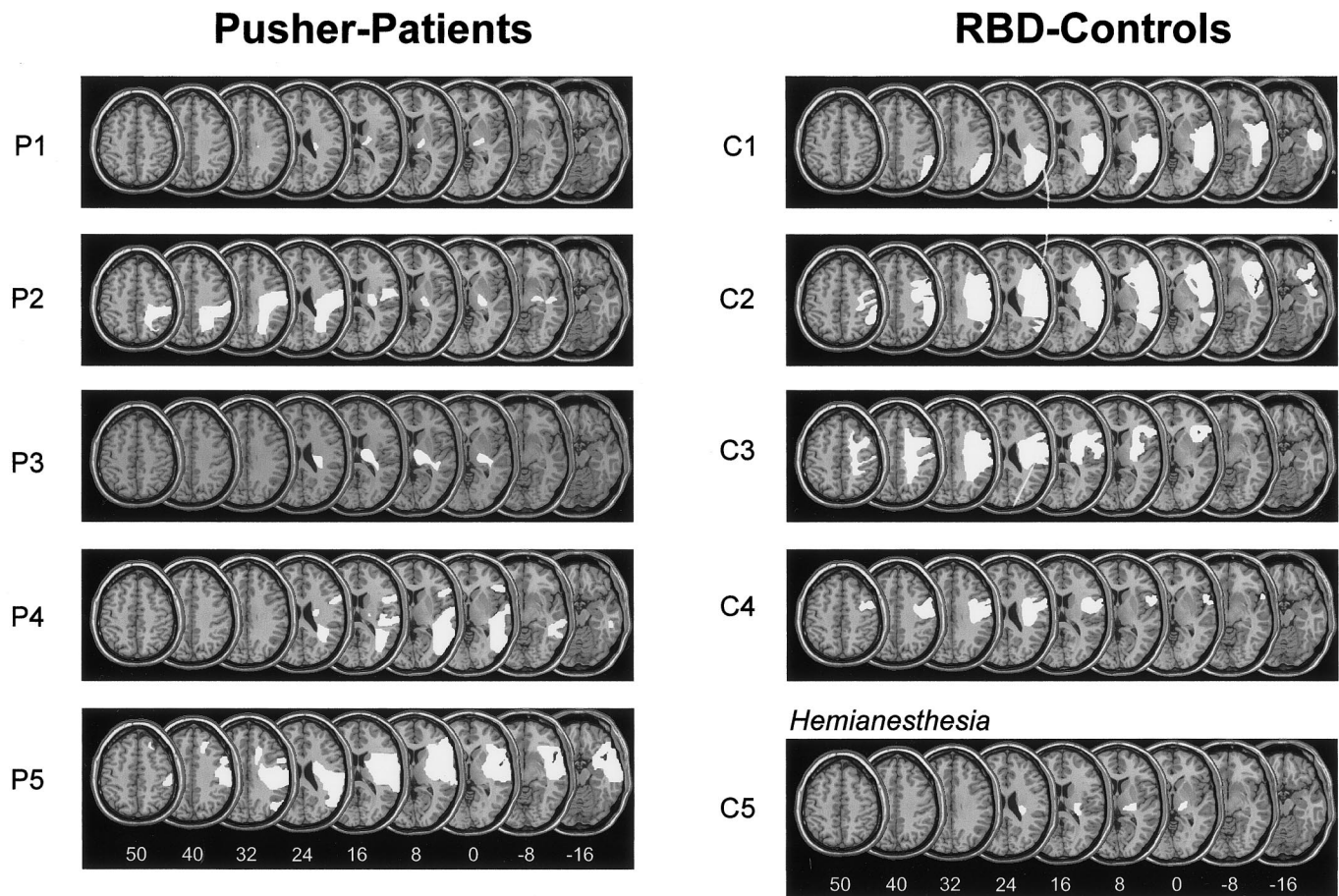


Figure 1. Infarcted areas of the patients with contraversive pushing (P1-5) and of the right brain-damaged (RBD) controls without pushing (C1-5). Brain lesions were identified by CT or MRI. Using MRIcro software (www.mrc-cbu.cam.ac.uk/~chris.rorden/micro.htm), the lesions were mapped on slices of a template MRI scan from the Montreal Neurologic Institute (www.bic.mni.mcgill.ca/cgi/icbm_view), which is based on 27 T1-weighted MRI scans of the same individual normalized to Talairach space.³² This scan is distributed with SPM99 (www.fil.ion.ucl.ac.uk/spm/spm99.html). The z coordinates of each transverse section are given.

0.3°, SD 0.8°). Accordingly, no apparent tilt of their SVV was found (mean -0.4° , SD 1.7°). These results corroborate earlier findings comparing the perception of the SPV and SVV in 40 patients with different brain lesions and in 20 non-brain-damaged subjects.²⁴

The pusher patients displayed an enormous tilt of the SPV toward the ipsilesional side without visual cues (see figure 2A). Their points of subjective verticality showed tilts of 21.8° (patient P1), 23.8° (P2), 14.8° (P3), 12.6° (P4), and 16.4° (P5). Each single position (as well as the mean of 17.9° , SD 4.7° , see figure 2A) was far beyond the 95% CI of the 10 control patients (-0.2° to 1.0°). This enormous ipsiversive tilt cannot be explained by spatial neglect or by a loss of somatosensory contact cues from the paretic body side, since the four brain-damaged controls with spatial neglect and the patient with complete hemianesthesia (C5) did not show such a tilt (C1, -0.2° ; C2, -0.5° ; C3, 1.8° ; C4, 0.2° ; C5: 0.3°).

In contrast, while viewing at the structured surroundings (see figure 2B), the pusher patients' point of subjective postural verticality (mean 0.9° , SD 1.6°) was within the 95% CI of the controls (-0.2° to 0.9°). Accordingly, their SVV (mean -0.4° , SD 2.5°) was within the controls' 95% CI (-1.7° to 0.8°).

In two of the pusher patients (P4 and P5), we conducted two additional experiments. One was to prove that the ability of the pusher patients to orient the SPV earth-vertical when allowed to look around in the laboratory (see figure 2B) was derived by visually aligning the longitudinal body axis earth-vertical with the upright orientation of the surrounding objects. As before, body rotations were performed with open eyes, but now the subjects were wearing Frenzel glasses and were facing a white plane. Under this condition, the patients' position of subjective verticality showed an average tilt of 16.6° (SD 0.1°), which was comparable with the tilt measured when their eyes had been occluded (15.7° , SD 4.3°).

In a further experiment, we again determined the two patients' perception of SVV in complete darkness. However, now they performed this task when they were tilted in the position that they subjectively perceived as "upright" (i.e., 15° (P4) and 17° (P5) to the right, averaged over the two conditions when their eyes were occluded and when they were wearing Frenzel glasses). We found an average SVV judgment of -1.4° (SD 2.0°), which was comparable with the SVV judgment when the two patients had been sitting physically upright (-1.1° , SD 1.9°).

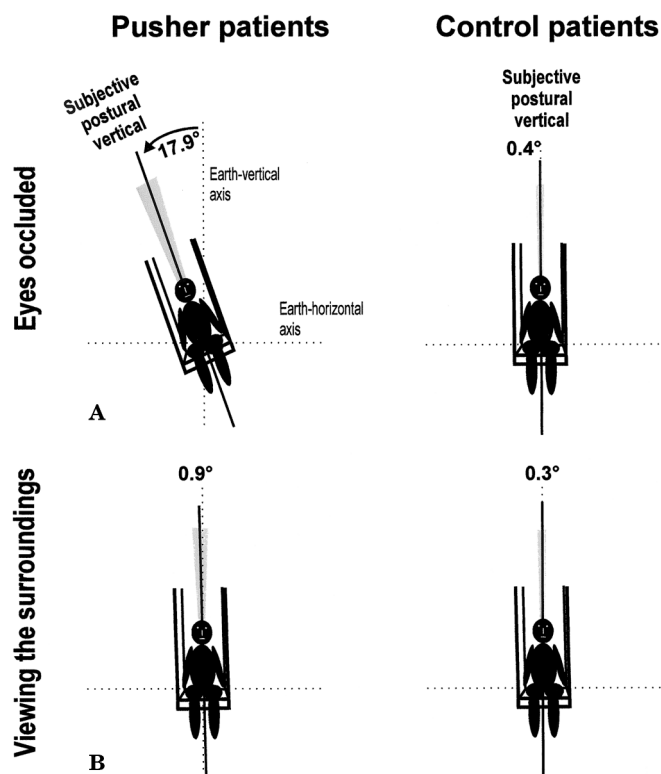


Figure 2. Schematic drawing of the subjects' perceived postural vertical with occluded eyes (A) and while viewing the laboratory's structured surroundings (B). The subjects are illustrated as seen from their front. The gray areas indicate the averaged sector of subjective verticality; the bold lines indicate the axis of subjective verticality.

Discussion. Why are the patients pushing toward the contralesional side? The current results show that the brain lesions in patients with contraversive pushing led to a severe ipsiversive tilt of perceived body verticality in the frontal plane. Without visual cues from the surroundings, which were achieved either by occluding the eyes or by wearing Frenzel glasses, the SPV was found tilted enormously toward the ipsilesional side, indicating that pusher patients experience their body's longitudinal axis oriented upright at this point. In contrast, by using visual cues from the laboratory's surroundings, the same patients can align their longitudinal body axis to earth-vertical. Also, without a visual surround in complete darkness, these patients can correctly determine visual vertical when sitting upright as well as when tilted in the position that is subjectively perceived as upright. The latter results indicate undisturbed processing of visual and vestibular inputs for the determination of visual vertical. Consequently, when pusher patients sit objectively upright, they experience a mismatch between visual vertical, based on vestibular and visual inputs on the one side, and the (tilted) orientation of subjective body verticality on the other.

Generally, a conflict between two reference systems is resolved either by suppressing one of them,

or both, or by a compromise (e.g., by weighted summation). However, neither happens in the current case. Under normal bedside conditions, the pusher patients do not align their body with the visual vertical, or with their perceived postural vertical, or with an intermediate posture. Instead, they move the body into the *opposite* direction. Although the current study cannot definitively clarify this discrepancy, it is speculated that by pushing their longitudinal body axis toward the contralesional side, the patients try to actively compensate for the mismatch between visual vertical and the (tilted) orientation of body verticality. Contraversive pushing did not occur when the pusher patients were tilted to the ipsilesional side (the subjective upright position) and structured visual input was excluded (by occluding the eyes or by wearing Frenzel glasses). Thus, the therapist's attempt to correct the patient's body posture toward the objective upright position (undertaken with eyes open) seems to contradict the patient's effort to "compensate" the mismatch between visual vertical and the (tilted) orientation of subjective body verticality, which induces the feeling of lateral instability and the fear of falling and provokes their actively resisting such attempts.

Alternatively, it is possible that the pushing behavior is a secondary response to the patients' unexpected experience that they lose lateral balance when trying to get up and sit upright in an upright room. The current experiment reveals that the patients' subjectively perceived upright orientation is tilted about 18° toward the ipsilesional side. Thus, when the patients try to get up and orient the body (subjectively) upright, they become laterally unstable because the center of mass is shifted too far to the ipsilesional side. Pushing the body to the opposite (contralesional) side might be the ensuing reaction to this experience. Hence, no pushing occurs when the patients sit immobilized by lateral stabilization in the cushioned safety of the experimental chair.

Future studies should further investigate these possible interpretations. Nevertheless, our experiment shows for the first time that the underlying cause of contraversive pushing may be a severe misperception of body orientation in relation to gravity.

Recent experiments^{6,7} have found a dissociation between the SVV and the SPV in patients with acute unilateral vestibular lesions. Whereas these patients showed a tilted SVV, their SPV remained earth-vertical. Here, we report the opposite behavioral dissociation in brain-damaged patients with contraversive pushing. Whereas our patients had an undisturbed SVV, their SPV showed a severe ipsiversive tilt. This double dissociation argues for different pathways subserving these two perceptions. (This notion is supported by earlier observations that the effect of optokinetic stimulation is smaller on the SPV than that on the SVV when normal subjects were required to maintain their upright orientation against random perturbations.²⁵)

How could these two perceptions or graviceptive systems be implemented in the brain? One possible assumption is that both systems rely on the same peripheral (visual, vestibular, eye- and neck-proprioceptive) input sources but that this same afferent input is projected to two anatomically separate neural networks that process the input in different ways. Whereas the first system processes the orientation of the visual world and the head to the vertical, the second system processes the posture of the trunk. If so, the current results indicate that both networks can be lesioned separately in different stroke patients, leading either to vestibular dysfunction in terms of a perceptual tilt of the visual vertical (as observed in patients with "vestibular cortex" lesions⁵) or to dyscontrol of body posture in terms of a perceptual tilt of the postural vertical (as observed in the current brain-damaged patients with contraversive pushing).

An alternative assumption would be that both graviceptive systems rely on (at least in part) *different* input sources. The latter has been suggested by Mittelstaedt.²⁶ He proposes that the orientation of the visual world and the head to the vertical is perceived exclusively through our (visual, vestibular, proprioceptive) sense organs in the head and the neck, whereas the posture of the trunk is mainly perceived through sense organs in the trunk. Such a truncal graviceptive system is known to exist in pigeons.²⁷⁻²⁹ As possible candidates for truncal graviceptors in humans, he assumed the afferent input from the kidneys through the renal nerve, and second, the information through the inertia of a mass in the body.²⁶ The blood in the large vessels or the mass of the abdominal viscera are suggested to contribute to the latter afferent input submitted through the phrenic or the vagus nerve.

In contrast, somatosensory input does not appear to contribute essentially to this second graviceptive system. Leg proprioceptors were found to have no direct access to the perception of posture.^{26,30} Proprioceptive input from the lower extremities appeared to affect the perception and control of posture only indirectly by implementing or modulating the output of the truncal graviceptors. The current results corroborate this notion on a minor contribution of somatosensory input for the perception of body posture. Our patient with complete left-sided sensory loss from a right thalamic lesion (patient C5) did not show any difficulties adjusting the SPV earth-vertical. (Indeed, the latter is a new observation on its own and supplements a recent study,³¹ which could obtain only SVV but no SPV estimates from two such patients with extreme sensory loss.)

Also, visual input does not seem to be a principal source of our perception of body upright. When visual and postural cues conflict, healthy subjects rely on postural cues rather than invalid visual information (see Mann et al.³). In the current study, when the pusher patients visually explored the structured surroundings, they were able to correctly align their

longitudinal body axis to earth-vertical. This indicates that seeing the upright orientation of the surrounding objects and persons helped their impaired graviceptive system. When vision of the surrounding contours was excluded by wearing Frenzel glasses and by facing a white plane, the patients again displayed the enormous tilt toward the ipsilesional side, which they had been shown with their eyes occluded. However, in contrast to these findings, pusher patients' spontaneous orientation of the body in daily life activities (i.e., in situations in which their eyes are open and they are viewing the surroundings) shows the characteristic tilt from their active pushing away from the nonparetic side. They even resist any attempt to passively correct their tilted body posture because they fear falling toward the non-paralyzed side. Thus, with the help of visual cues, these patients can only temporarily align their body axis to earth-vertical under certain constrained laboratory conditions in which the body was immobilized by lateral stabilization, the legs were freely hanging, and the patients were explicitly instructed to concentrate on visual information taken from the environment. Apparently, visual input does not suffice to control continuously upright body posture in pusher patients.

Nevertheless, it may be speculated that for rehabilitation, this preserved ability to align the body axis to earth-vertical with the help of visual cues might be a helpful new tool. Although the patients are not spontaneously able to use the visual input to control upright body posture, this might become possible when training procedures apply this ability for conscious strategies of posture control in these subjects.

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